

Problem 1

Use the Cauchy-Riemann conditions to find out whether the functions in (1)~(4) are analytic.

(1) $\frac{2z-i}{iz+2}$

assume $z=x+iy$, $f(z)=\frac{2z-i}{iz+2}$

$$\frac{\partial f}{\partial x} = \frac{df}{dz} \cdot \frac{\partial z}{\partial x} = \frac{2(iz+2)-(2z-i) \cdot i}{(iz+2)^2} \cdot 1$$
$$= \frac{3}{(iz+2)^2}$$

$$\frac{\partial f}{\partial y} = \frac{df}{dz} \cdot \frac{\partial z}{\partial y} = \frac{3i}{(iz+2)^2}$$

$$i \frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} \Rightarrow \text{Analytic, but } z \neq -2i$$

$$f = u + iv$$

Cauchy-Riemann Condition

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\frac{\partial f}{\partial x} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$\frac{\partial f}{\partial y} = \frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y}$$

$$i \frac{\partial f}{\partial x} = -\frac{\partial v}{\partial x} + i \frac{\partial u}{\partial x}$$

Equivalence of C-R:

$$i \frac{\partial f}{\partial x} = \frac{\partial f}{\partial y}$$

(2) $z^2 \bar{z}$

assume $z=x+iy$, $f(z)=z^2 \bar{z}$

$$f(x,y) = (x+iy)^2 (x-iy) = x^3 + xy^2 + i(x^2y + y^3)$$

$$\frac{\partial u}{\partial x} = 3x^2 + y^2, \quad \frac{\partial v}{\partial y} = x^2 + 3y^2$$

$$\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y} \Rightarrow \text{Not analytic}$$

(3) $\sqrt{z}$

assume $z=x+iy$, $f(z)=\sqrt{z}$

$$\frac{\partial f}{\partial x} = \frac{df}{dz} \frac{\partial z}{\partial x} = \frac{1}{2\sqrt{z}}, \quad \frac{\partial f}{\partial y} = \frac{df}{dz} \frac{\partial z}{\partial y} = \frac{i}{2\sqrt{z}}$$

$$i \frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} \Rightarrow \text{Analytic, but } z \neq 0$$

$$(4) e^{iz}$$

$$\text{assume } z = x + iy, f(z) = e^{iz}$$

$$\frac{\partial f}{\partial x} = \frac{df}{dz} \frac{\partial z}{\partial x} = i e^{iz}, \quad \frac{\partial f}{\partial y} = \frac{df}{dz} \frac{\partial z}{\partial y} = -e^{iz}$$

$$i \frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} \Rightarrow \text{Analytic}$$

Problem 2

Show that the following functions are harmonic, that is, that they satisfy Laplace's equation, and find for each a function $f(z)$ of which the given function is the real part. Show that the function $v(x, y)$ (which you find) also satisfies Laplace's equation.

$$(1) 3x^2y - y^3$$

$$\text{assume } u(x, y) = 3x^2y - y^3$$

$$\frac{\partial^2 u}{\partial x^2} = 6y, \quad \frac{\partial^2 u}{\partial y^2} = -6y \Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \Rightarrow u(x, y) \text{ is harmonic}$$

Cauchy - Riemann condition:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 6xy, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = 3x^2 - 3y^2$$

$$\Rightarrow \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 v}{\partial y^2} = 6x, \quad \frac{\partial^2 u}{\partial y \partial x} = -\frac{\partial^2 v}{\partial x^2} = 6x$$

$$\Rightarrow \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 \Rightarrow v(x, y) \text{ is harmonic}$$

$$(2) e^x \cos y$$

$$\text{assume } u(x, y) = e^x \cos y$$

$$\frac{\partial^2 u}{\partial x^2} = e^x \cos y, \quad \frac{\partial^2 u}{\partial y^2} = -e^x \cos y \Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\Rightarrow u(x, y) \text{ is harmonic}$$

Cauchy - Riemann condition :

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = e^x \cos y, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = -e^x \sin y$$

$$\Rightarrow \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 v}{\partial y^2} = -e^x \sin y, \quad \frac{\partial^2 u}{\partial y \partial x} = -\frac{\partial^2 v}{\partial x^2} = -e^x \sin y$$

$$\Rightarrow \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 \Rightarrow v(x, y) \text{ is harmonic}$$

(3) $\ln(x^2 + y^2)$

assume $u(x, y) = \ln(x^2 + y^2)$

$$\frac{\partial u}{\partial x} = \frac{2x}{x^2 + y^2}, \quad \frac{\partial^2 u}{\partial x^2} = \frac{2(y^2 - x^2)}{(x^2 + y^2)^2}$$

$$\frac{\partial u}{\partial y} = \frac{2y}{x^2 + y^2}, \quad \frac{\partial^2 u}{\partial y^2} = \frac{2(x^2 - y^2)}{(x^2 + y^2)^2}$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \Rightarrow u(x, y) \text{ is harmonic}$$

Cauchy - Riemann condition :

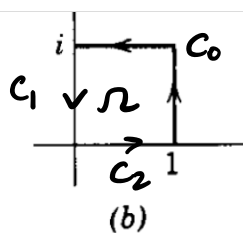
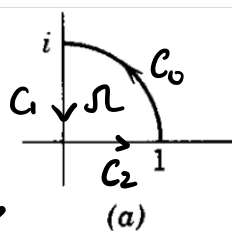
$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = \frac{2x}{x^2 + y^2}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = \frac{2y}{x^2 + y^2}$$

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y^2} = \frac{4xy}{(x^2 + y^2)^2}, \quad \frac{\partial^2 u}{\partial y \partial x} = -\frac{\partial^2 v}{\partial x^2} = \frac{4xy}{(x^2 + y^2)^2}$$

$$\Rightarrow \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 \Rightarrow v(x, y) \text{ is harmonic}$$

Problem 3

$\int_1^i z dz$ along the indicated paths:



$f(z)=z$ is analytic in region Ω

Cauchy's theorem

$$\oint_C z dz = \int_{C_0} z dz + \int_{C_1} z dz + \int_{C_2} z dz = 0$$

$$\begin{aligned} \int_{C_0} z dz &= -\int_{C_1} z dz - \int_{C_2} z dz & z=x+iy, \quad dz=d(x+iy) \\ &= -\int_1^0 iy diy - \int_0^1 x dx \\ &= -\frac{1}{2} - \frac{1}{2} = -1 \end{aligned}$$

Problem 4

If $f(z)$ is analytic on and inside the circle $|z|=1$, show that $\int_0^{2\pi} e^{i\theta} f(e^{i\theta}) d\theta = 0$.

$$\text{Cauchy's theorem: } \oint_{|z|=1} f(z) dz = 0$$

$$\Rightarrow \int_0^{2\pi} f(e^{i\theta}) de^{i\theta} = 0$$

$$\int_0^{2\pi} ie^{i\theta} f(e^{i\theta}) d\theta = 0$$

$$\Rightarrow \int_0^{2\pi} e^{i\theta} f(e^{i\theta}) d\theta = 0$$

Problem 5

Use Cauchy's theorem or integral formula to evaluate the integrals

$$\oint_C \frac{\sin z \, dz}{2z - \pi} \text{ where } C \text{ is the circle } \begin{matrix} \text{(a) } |z| = 1, \\ \text{(b) } |z| = 2. \end{matrix}$$

assume $f(z) = \frac{\sin z}{2z - \pi}$, $f(z)$ is not analytic when $z = \frac{\pi}{2}$

Cauchy's theorem: $\oint_{|z|=1} \frac{\sin z}{2z - \pi} dz = 0$

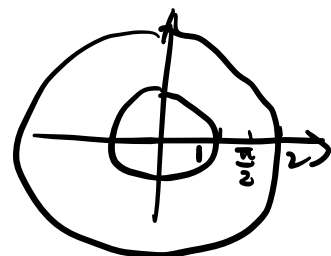
$$\oint_{|z|=2} \frac{\sin z}{2z - \pi} dz = \oint_{|z|=2} \frac{\frac{1}{2} \sin z}{z - \frac{\pi}{2}} dz$$

assume $g(z) = \frac{1}{2} \sin z$, $g(z)$ is analytic

Cauchy's integral formula:

$$\oint_{|z|=2} \frac{g(z)}{z - \frac{\pi}{2}} dz = 2\pi i g\left(\frac{\pi}{2}\right) = i\pi$$

$$\Rightarrow \text{(a) } 0 \quad \text{(b) } i\pi$$



Problem 6

Use Cauchy's theorem or integral formula to evaluate the integrals

$$\oint_C \frac{\sin 2z \, dz}{6z - \pi} \text{ where } C \text{ is the circle } |z| = 3.$$

$$\oint_C \frac{\frac{1}{6} \sin 2z}{z - \frac{\pi}{6}} dz \quad \text{assume } f(z) = \frac{1}{6} \sin 2z, f(z) \text{ is analytic in } |z| = 3$$

$$\Rightarrow \oint_C \frac{f(z)}{z - \frac{\pi}{6}} dz = 2\pi i f\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{6} \pi i$$

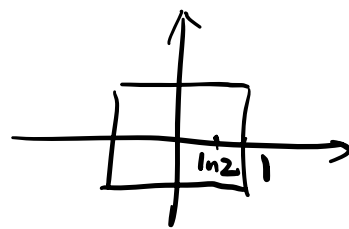
Problem 7

Use Cauchy's theorem or integral formula to evaluate the integrals

$$\oint \frac{e^{3z} dz}{z - \ln 2} \text{ if } C \text{ is the square with vertices } \pm 1 \pm i.$$

assume $f(z) = e^{3z}$. $f(z)$ is analytic in the region

$$\Rightarrow \oint_C \frac{f(z)}{z - \ln 2} dz = 2\pi i f(\ln 2) = 16\pi i$$



Problem 8

$$\begin{aligned} (1) \quad \oint_{|z|=2} \frac{z^2-1}{z^2+1} dz &= \oint_{|z|=2} (z^2-1) \frac{1}{2i} \left(\frac{1}{z-i} - \frac{1}{z+i} \right) dz \\ &= \oint_{|z|=2} \frac{-\frac{i}{2}(z^2-1)}{z-i} dz - \oint_{|z|=2} \frac{-\frac{i}{2}(z^2-1)}{z+i} dz \end{aligned}$$

assume $f(z) = -\frac{i}{2}(z^2-1)$, $f(z)$ is analytic in $|z|=2$

$$\Rightarrow \oint_{|z|=2} \frac{z^2-1}{z^2+1} dz = 2\pi i f(i) - 2\pi i f(-i) = 0$$

$$(2) \quad \oint_{|z|=2} \frac{\sin(e^z)}{z} dz = 2\pi i \cdot \sin(e^0) = i 2\pi \sin 1$$

$$(3) \quad \oint_{|z|=2} \frac{\sin z}{z^4} dz$$

assume $f(z) = \sin z$, $f(z)$ is analytic in $|z|=2$

$$f^{(3)}(0) = \frac{3!}{2\pi i} \oint_{|z|=2} \frac{\sin z}{z^4} dz$$

$$\Rightarrow \oint_{|z|=2} \frac{\sin z}{z^4} dz = -\frac{\pi}{3} i$$

$$(4) \oint_{|z|=2} \frac{dz}{z^2(z^2+16)} = \oint_{|z|=2} \frac{\frac{1}{z^2+16}}{(z-0)^2} dz$$

assume $f(z) = \frac{1}{z^2+16}$, $f(z)$ is analytic in $|z|=2$

$$f'(0) = \frac{1}{2\pi i} \oint_{|z|=2} \frac{f(z)}{(z-0)^2} dz = \left. \frac{-2z}{(z^2+16)^2} \right|_{z=0} = 0$$

$$\Rightarrow \oint_{|z|=2} \frac{dz}{z^2(z^2+16)} = 0$$